

BookScope API Documentation

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Overview

BookScope API is a REST API for book catalog browsing, personal bookshelf management, reviews, explainable recommendations, and reading analytics.

Authentication

Authentication uses Bearer tokens via JSON Web Tokens.

Register

- `POST /api/auth/register/`

Example request:

```
{
  "username": "reader1",
  "email": "reader1@example.com",
  "password": "StrongPass123!"
}
```

Obtain Access Token

- `POST /api/auth/token/`

Example request:

```
{
  "username": "reader1",
  "password": "StrongPass123!"
}
```

Refresh Token

- `POST /api/auth/token/refresh/`

Current User

- `GET /api/auth/me/`

Requires:

- `Authorization: Bearer `

Catalog Endpoints

Books

- `GET /api/books/`
- `POST /api/books/`
- `GET /api/books/{id}/`
- `PATCH /api/books/{id}/`
- `DELETE /api/books/{id}/`

Supported list query parameters:

- `search`
- `genre`
- `author`
- `language`
- `year\_min`
- `year\_max`
- `ordering`
- `page`
- `page\_size`

Authors

- `GET /api/authors/`
- `POST /api/authors/`
- `GET /api/authors/{id}/`
- `PATCH /api/authors/{id}/`
- `DELETE /api/authors/{id}/`

Genres

- `GET /api/genres/`
- `POST /api/genres/`
- `GET /api/genres/{id}/`
- `PATCH /api/genres/{id}/`
- `DELETE /api/genres/{id}/`

Bookshelf Endpoints

These endpoints require authentication.

- `GET /api/me/bookshelf/`
- `POST /api/me/bookshelf/`
- `GET /api/me/bookshelf/{id}/`
- `PATCH /api/me/bookshelf/{id}/`
- `DELETE /api/me/bookshelf/{id}/`

Example request:

```
{
  "book_id": 1,
  "status": "reading",
  "personal_rating": 5,
  "is_favorite": true,
  "notes": "Strong first impression."
}
```

Review Endpoints

- `GET /api/books/{id}/reviews/`
- `POST /api/books/{id}/reviews/`
- `PATCH /api/reviews/{id}/`
- `DELETE /api/reviews/{id}/`

Example request:

```
{
  "rating": 5,
  "title": "Excellent",
  "content": "Focused, practical, and easy to apply.",
  "contains_spoiler": false
}
```

Recommendation Endpoints

Personalized Recommendations

- `GET /api/recommendations/for-me/`

Behavior:

- uses personal bookshelf ratings and review ratings
- excludes books already present in the user's bookshelf or review history

- returns human-readable recommendation reasons

Similar Books

- ``GET /api/recommendations/similar-books/{book_id}/``

Analytics Endpoints

- ``GET /api/analytics/books/trending/``
- ``GET /api/analytics/genres/popularity/``
- ``GET /api/analytics/ratings/distribution/``
- ``GET /api/analytics/me/reading-summary/``
- ``GET /api/analytics/authors/top/``

Error Handling

The API returns structured JSON errors.

Example response:

```
{
  "error": {
    "code": "validation_error",
    "message": "Validation failed.",
    "details": {
      "book_id": [
        "You already have a bookshelf entry for this book."
      ]
    }
  }
}
```

Status Code Conventions

- ``200 OK``
- ``201 Created``
- ``204 No Content``
- ``400 Bad Request``
- ``401 Unauthorized``
- ``403 Forbidden``
- ``404 Not Found``

Google Books Data Import

The catalog is designed to ingest Google Books metadata into the local SQLite database.

Typical workflow:

1. Fetch Google Books payloads.
2. Normalize titles, authors, genres, ISBNs, and metadata.
3. Store them in local tables.
4. Build recommendations and analytics from local data plus user activity.

OpenAPI Schema

The machine-readable schema is exported to ``docs/api_openapi.yaml``.